

Disguise feature	Retrieval	method	Qwen-max	Llama3.3-70B-instruct	deepseek-v3
✓	✓	RADAR (NIPS'23)	34.00	62.50	22.00
		DeTeCtive (NIPS'24)	81.00	91.50	75.50
		MPU(ICLR'24)	33.00	21.50	6.50
		chatgpt detector	21.50	11.50	3.00
		Average	42.38	46.75	26.75
✗	✓	RADAR (NIPS'23)	40.50	46.50	32.00
		DeTeCtive (NIPS'24)	83.50	88.00	71.00
		MPU(ICLR'24)	38.50	50.50	6.00
		chatgpt detector	28.00	29.00	2.00
		Average	47.63	53.50	27.75
✓	✗	RADAR(NIPS'23)	49.50	99.50	45.50
		DeTeCtive (NIPS'24)	61.50	90.50	66.50
		MPU(ICLR'24)	48.00	14.50	0.00
		chatgpt detector	21.00	1.00	0.00
		Average	45.00	51.38	28.00
✗	✗	RADAR (NIPS'23)	36.50	48.00	56.50
		DeTeCtive (NIPS'24)	98.50	100.00	90.00
		MPU(ICLR'24)	80.50	84.50	39.00
		chatgpt detector	27.00	38.50	10.50
		Average	60.63	61.50	49.00

Table 4: Ablation study. "Disguise feature" indicates whether the features extracted by the adversarial feature extractor are used. "Retrieval" denotes whether the context examples optimizer is employed. The "✓" indicates that the component is enabled, while "✗" that the component is not employed. The optimal results are highlighted in bold in the table.

Efficiency Experiments

SDA does not require fine-tuning the model, thereby reducing computational resources during training. However, considering the impact of longer prompts on generation time, we compare the time required to generate 50 samples using Qwen-max. As shown in Figure 4, SDA introduces only minimal overhead during inference compared to SICO, demonstrating its efficiency.

Ablation Study

We validate the effectiveness of the two main components, the adversarial feature extractor and the retrieval-based context examples optimizer, through ablation experiments. The test results for different combinations are shown in the table 4. Under the guidance of the proposed SDA, the average detection accuracy of texts generated by the three latest LLMs decreased by more than 14%. Additionally, the experimental results show that, incorporating the features extracted by the feature extractor significantly reduces the average detection accuracy. Specifically, compared to direct generation, the average detection accuracy decreases by over 10% across all three LLMs when disguise features are included. Furthermore, the decrease in consistency across the three LLMs suggests that the disguise features generated by one LLM can generalize to others, indicating an inherent correlation between the texts produced by different LLMs. Moreover, incorporating a retrieval-based context examples optimizer to refine the selection of context examples can also significantly reduce the average detection accuracy of model. We

also conduct additional ablation experiments, including introducing easily detectable samples during the feature extraction and selecting different proxy detectors, to validate the rationality of our approach. The experimental results are shown in Appendix E. Notably, we propose a potential defense solution against SDA, as detailed in Appendix F.

Conclusion

In conclusion, we propose a novel detection evasion method SDA that effectively guides the LLM to disguise its output in order to generate more human-like text. The SDA primarily consists of two components: the adversarial feature extractor and retrieval-based context examples optimizer. The former is designed to capture the key disguise features that enable the LLM to understand how to generate human-like text. The latter improves the selection of context examples, further enhancing the self-disguise ability of LLMs and mitigating the impact of the disguise process on the diversity of the generated text. Finally, we construct a self-disguise generation prompt combined with optimized contextual examples and disguise features to guide the target LLM in generating detection-resistant texts. Our extensive experiments on evasion demonstrate the superior performance of SDA, which significantly reduces the detection probability of AIGT while effectively preserving text quality.

Ethics Statement

The primary objective of this paper is not to provide techniques for evading AIGT detection systems, but rather to

expose vulnerabilities in current detection mechanisms. We firmly believe that with increased attention and efforts toward this issue, the research community can devise more sophisticated and effective techniques to enhance the robustness and reliability of machine-generated text detection systems in the face of evolving adversarial threats.

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